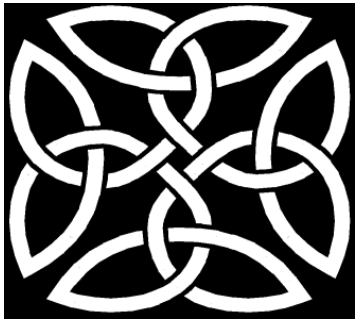




The University of Edinburgh

College of Science and Engineering
School of Mathematics

Mathematics 3 Honours
MATH10003 Financial Mathematics

Wednesday 11th December 2019

9.30am-11.30am

Chairman of Examiners – Professor A Olde Daalhuis
External Examiner – Professor J McColl

Answer **ALL** questions

Calculators and other electronic aids:

A scientific calculator is permitted in this examination.

It must not be a graphical calculator.

It must not be able to communicate with any other devices.

This examination will be marked anonymously.

- (1) Consider a 3-period binomial model over the next year (three 4-months periods) with a stock S . For each period the stock price goes up by 25% with probability 0.15 or down by 25% with probability 0.85. Assume that $S_0 = 400$ and the risk-free interest is at a rate of $r = 5\%$ discretely per period. Correct all numbers up to two decimal places after the decimal point.
- (a) Draw a tree to show all trajectories of S and calculate the risk-neutral probability q . **[2 marks]**
- (b) Consider an European up and out barrier call option with a payoff:

$$h = \begin{cases} S_3 - 460.71, & \text{if } S_n < 603.72 \text{ for all } n = 0, 1, 2, 3, \\ 0, & \text{if } S_n \geq 603.72 \text{ for some } n = 0, 1, 2, 3. \end{cases}$$

- Determine the value of this option *along each trajectory*. You can add the values to each node of the tree (in brackets for example). **[5 marks]**
- (c) Determine the value of the American counterpart of the up and out call option stated above. Write down the optimal exercise date. (Hint: the barrier condition only affects the time before the exercise date.) **[7 marks]**

- (2) A chooser option on an underlying stock S with maturity T and strike K is a path dependent option that gives the buyer the choice of exercising either a European call option or a European put option on the same stock S with the same maturity T and the same strike price K . In this question, we consider that the buyer will make the choice at a fixed time T_0 , $0 < T_0 < T$, at which time the buyer will choose the option with higher value at T_0 .

Define also the event

$$A = \{ \Pi(T_0; \text{Call}(T, K)) < \Pi(T_0; \text{Put}(T, K)) \},$$

where $\Pi(T_0; \text{Call}(T, K))$ and $\Pi(T_0; \text{Put}(T, K))$ denotes the price of call and put option at time T_0 respectively with maturity T and strike price K .

- (a) Show that the payoff function of the chooser option at maturity T can be expressed as

$$\max((S_T - K), 0) + (K - S_T)\mathbf{1}\{A\}.$$

[3 marks]

- (b) Consider the call and put option on the stock S , with maturity T and strike K . State the corresponding put call parity at time T_0 , where $0 < T_0 < T$.

[3 marks]

- (c) Assuming that the market is arbitrage-free and the interest rate is r , continuously compounded. Using part (a) and (b), show that at time 0, the premium of the chooser option with maturity T and strike K is equivalent to the premium of the call with maturity T and strike K plus the premium of the put option with maturity T_0 and strike $Ke^{-r(T-T_0)}$, in other words,

$$\Pi(0; \text{Chooser}(T, K)) = \Pi(0; \text{Call}(T, K)) + \Pi(0; \text{Put}(T_0, Ke^{-r(T-T_0)})).$$

(Hint: You may use the fact that under event A , we have $S_T = S_{T_0}e^{r(T-T_0)}$.)

[7 marks]

- (3) (a) State the Itô's formula for a smooth function $f(t, W_t)$ of time t and the standard Brownian motion $(W_t)_{t \geq 0}$ in integral form. **[4 marks]**

- (b) Let $\mu, \sigma \in \mathbb{R}$ with $\sigma > 0$. For $t > 0$ we consider

$$X_t = e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t}.$$

Use Itô's formula to find a stochastic differential equation for X_t in terms of dt, dW_t, μ and σ . **[5 marks]**

- (c) Determine the range of values of μ and σ such that for any $t > 0$ we have

$$\mathbb{E}[X_t] = 1.$$

[3 marks]

- (4) Consider $T > 0$, $t \in [0, T]$ and let $\mu, \sigma \in \mathbb{R}$. Let $(X_s)_{s \in [t, T]}$ be the Itô process defined as

$$\begin{aligned} dX_s &= (-X_s - X_s^3) dt + X_s dW_s, \quad \text{for } s \in [t, T] \\ X_t &= x. \end{aligned}$$

Assume that $f \in C^{1,2}([0, T] \times \mathbb{R}, \mathbb{R})$ is a solution to the following boundary value problem partial differential equation for $(t, x) \in [0, T] \times \mathbb{R}$,

$$\partial_t f(t, x) - (x + x^3) \partial_x f(t, x) + \frac{1}{2} x^2 \partial_{xx} f(t, x) - r f(t, x) = 0,$$

and boundary condition $f(T, x) = 2x^2$ for $x \in \mathbb{R}$.

- (a) Show that

$$e^{-rt} f(t, X_t) = e^{-rT} f(T, X_T) - \int_t^T [e^{-rs} X_s \partial_x f(s, X_s)] dW_s.$$

[7 marks]

- (b) Assume that the process $(e^{-rs} X_s \partial_x f(s, X_s))_{s \in [t, T]} \in L^2([t, T])$. Prove that for any $(t, x) \in [0, T] \times \mathbb{R}$ the value $f(t, x)$ can be expressed by

$$f(t, x) = 2e^{-r(T-t)} \mathbb{E}[X_T^2].$$

[4 marks]